

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Pharm. Examination

University Theory Examination April 2016

PH217 Pharmaceutical Chemistry - III

Date: 13.04.2016 , Wednesday Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. **With respect to the electrophilic aromatic substitution of benzene which of the following is not true?**
[A] A non-aromatic intermediate is formed
[B] Benzene acts as an electrophile
[C] A proton is lost in the final step
[D] Resonance forms are important
2. **When considering electrophilic aromatic substitution reactions electron withdrawing substituents (e.g. nitro) are described as...**
[A] *Ortho/para* directing and activating
[B] *Ortho/para* directing and deactivating
[C] *Meta* directing and activating
[D] *Meta* directing and deactivating
3. **When considering electrophilic aromatic substitution reactions the halides are described as...**
[A] *Ortho/para* directing and activating
[B] *Ortho/para* directing and deactivating
[C] *Meta* directing and activating
[D] *Meta* directing and deactivating

4. Following are the facts about S_EAr Mechanism; EXCEPT ONE

- [A] It is two step reaction
- [B] Electron donating group on benzene ring increase the rate of reaction
- [C] Electron withdrawing group on benzene ring increase the rate of reaction
- [D] First step in mechanism is Rate determining step

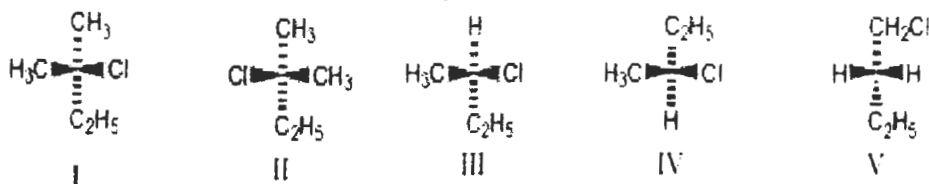
5. Which of the following statements is incorrect: aromatic compounds...

- [A] Are planar
- [B] Have $4n$ π -electrons
- [C] Are cyclic
- [D] Are generally less reactive than similarly substituted alkenes

6. Nucleophilic addition/substitution reaction is fast with _____

- [A] Ester
- [B] Ether
- [C] Amine
- [D] Alkane

7. Chiral molecules are represented by:



- [A] I, II, III, IV and V
- [B] I, II, III and IV
- [C] I and II
- [D] III and IV

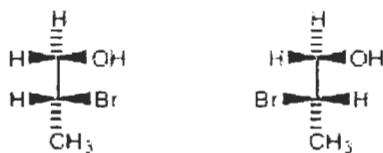
8. Which of the following is consider as Optical Isomers?

- [A] D and L
- [B] d and l
- [C] R and S
- [D] E and Z

9. Find out one of the correct statement from the options;

- [A] Aldehyde and Ketone gives substitution type of reaction
- [B] Nucleophilic substitution reaction is more faster at acyl carbon than saturated carbon
- [C] Ether is more reactive towards nucleophilic attack than ester
- [D] Amines is more reactive towards nucleophilic attack than amide

10. The molecules shown are:



- [A] Constitutional isomers
- [B] Enantiomers
- [C] Diastereomers
- [D] Identical

11. Benzene gives _____ types of reaction

- [A] Addition
- [B] Substitution
- [C] Elimination
- [D] All of above

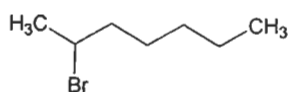
12. Following are the characteristics of the carboxylic acids. Except One

- [A] Dimer formation takes place
- [B] Electron withdrawing group increase the acidity of it
- [C] Polar in nature
- [D] Carboxylic acids have lower boiling point than alcohol

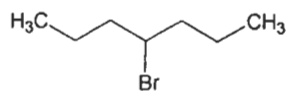
13. Which of the below structure is more stable than eclipsed

- [A] Staggered
- [B] Anti
- [C] Gauche
- [D] All of above

14. Structure A and B is a class of _____ isomer



A



B

- [A] Functional
- [B] Chain
- [C] Positional
- [D] Metamerism

15. Cope rearrangement is an example of _____ type of reaction.

- [A] [3,3'] Sigmatropic Rearrangement
- [B] [1,3] Sigmatropic Rearrangement
- [C] [1,5] Sigmatropic Rearrangement
- [D] [5,5'] Sigmatropic Rearrangement

16. With Odd no of π Bond, The Cycloaddition Reaction under Photochemical Condition in _____ fashion.

- [A] Conrotatory
- [B] Disrotatory
- [C] Antarafacial
- [D] Suprafacial

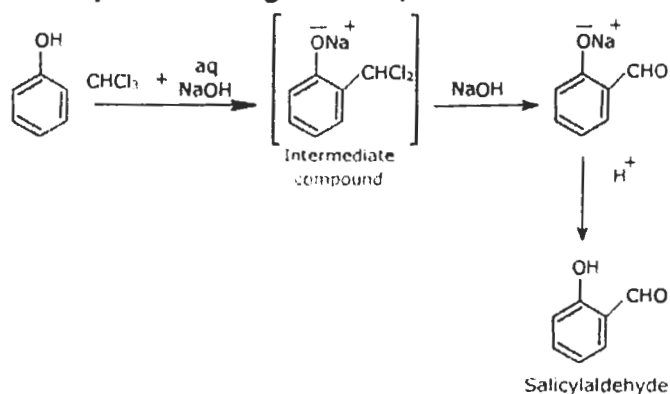
17. The term Atropisomerism is used for.....

- [A] That can be interconverted by rotation about single bonds
- [B] That are geometrical isomer
- [C] That are optical isomer
- [D] That are enantiomer

18. All are an examples of Heterogenous catalyst; EXCEPT One

- [A] Nickel
- [B] Platinum
- [C] Palladium
- [D] Wilkinson's

19. Identify the following reactions;



- [A] Wolff kishner Reduction
- [B] Aldol Condensation
- [C] Reimer Tiemann Reaction
- [D] Clemmensen Reduction

20. Hydrazine Hydrate reagent is used in _____ type of reaction.

- [A] Wolff kishner Reduction
- [B] Aldol Condensation
- [C] Reimer Tiemann Reaction
- [D] Clemmensen Reduction

SECTION – II

- Q 2** Attempt any **FOUR** of the following
- | | | |
|----------|--|-----------|
| A | Write any Three Preparations of Aldehyde. | 05 |
| B | Write any Three Preparations of Amide and Ester. | 05 |
| C | Explain Wolff Kishner Synthesis with suitable reaction and mechanism | 05 |
| D | Explain Reimer Tiemann Reaction with suitable reaction and mechanism | 05 |
| E | Explain Malonic ester synthesis and Williamson's Synthesis with suitable reaction and mechanism. | 05 |
| F | Write a Short Note on Sigmatropic Reaction. | 05 |

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- | | | |
|----------|--|-----------|
| A | Define the following terms; (Each definition is of one mark) | 05 |
| | i. Optical activity | |
| | ii. Eclipsed and Staggered | |
| | iii. Cis and Trans | |
| | iv. Racemic mixture | |
| | v. Geometrical isomerism | |
| B | Differentiates between the following with suitable examples. | |
| | i. Enantiomers and Diastereomers (03 Marks) | 05 |
| | ii. Configuration and Conformation (02 Marks) | |
| C | Explain Criteria of Aromaticity. | 05 |
| D | Write a Short Note on Cycloaddition Reaction. | 05 |
| E | Give preparations of Naphthalene. | 05 |
| F | Write a short note on Neighbouring Group Effect. | 05 |
- Q 4** Attempt any **FOUR** of the following
- | | | |
|----------|---|-----------|
| A | Explain nitration reaction of Benzene with mechanism. | 05 |
| B | What is Conformation? Explain Conformation in n-Butane with energy diagramme. | 05 |
| C | Write a note on Stereochemistry of Biphenyls. | 05 |
| D | Short note on Orientation in Mono substituted Benzene | 05 |
| E | Explain Reaction and Mechanism Friedel–Crafts Acylation. | 05 |
| F | Discuss the Sequence rule in detail. | 05 |
-